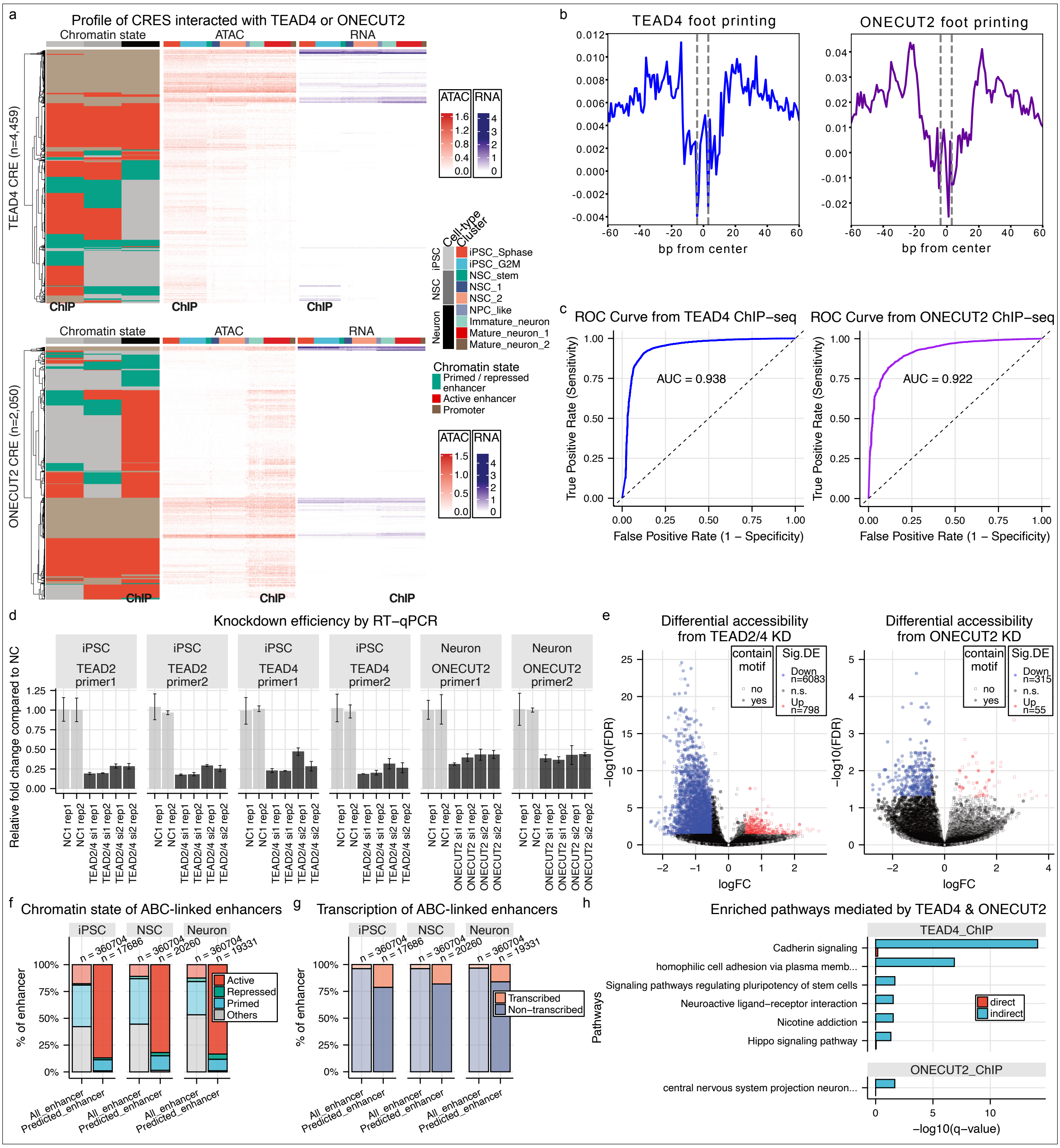

Extended Figure 4 | Validation of TEAD4 and ONECUT2 regulation via ChIP-seq and knockdown-ATAC-seq.

a, Heatmaps showing the chromatin states from the three samples, and ATAC and TSS signals from the metacells of the CREs interacted with TEAD4 (upper) and ONECUT2 (lower). Interacting CREs were defined by significant ChIP-seq and presence of the known motifs (TEAD[1-4] for TEAD4, ONECUT[1-3] for ONECUT2). **b**, Corrected ATAC signal of footprinting-positive CREs identified by TOBIAS. Results from TEAD4 from the cluster iPSC_Sphase and ONECUT2 from the cluster Immature_neuron are shown as representatives. **c**, ROC curves evaluating TOBIAS footprinting scores against ChIP-seq ground truth. Results from the cluster iPSC_Sphase and the cluster Immature_neuron were shown for TEAD4 and ONECUT2 respectively. **d**, RT-qPCR validation of siRNA knockdown of TEAD2/4 (iPSC) and ONECUT2 (neuron) using two independent primer sets for each gene. Relative expression levels comparing to the non-targeting control (NC1) siRNA are shown. **e**, MA plots of differential chromatin accessibility comparing the TF knockdown versus non-targeting control across defined CREs (with at least 100 counts from each replicate). Black dots represent non-significant CREs, filled circles indicate CREs containing specific TF motifs, while hollow squares represent CREs without the motifs. **f**, Chromatin states of all enhancers and enhancers that were predicted to link to other genes by ABC model (ABC score > 0.02) in three cell types. **g**, Transcription state of enhancers stratified as in (**f**). **h**, Pathway enrichment of genes directly and indirectly connected with TEAD4 in iPSC and ONECUT2 in neuron according to CREs identified from ChIP-seq datasets. For indirect enrichment, expressing genes linked to enhancers by ABC model (ABC score > 0.02) were used. Gene Ontology (GO) enrichment was performed using topGO with the elim algorithm while KEGG and Reactome pathway enrichment analyses were performed using clusterProfiler. Only significant pathways (q-value < 0.05) are shown.